TASK REPORT 01 · INFRASTRUCTURE & SYSTEMS MIGRATION

Full Cutover from GCP to Computer 4

Complete Technical Record of System Consolidation, State Rescue, Database Migration, and Cloud Decommissioning (October 8 – 16, 2026)

DATE OF CUTOVER	HARDWARE HOST	CLOUD STATUS	DATABASE ROWS
2026-10-16	Acer Nitro 60 (Win 11)	GCP Deleted & Closed	21,712,916 (0 Diff)

1. Motivation & Context

Prior to October 2026, Scalable Brain was fractured across three disparate hosting locations:

- **Computer 1:** Ubuntu Linux desktop executing System 1 batch cron jobs and TimescaleDB.
- **GCP VM `trading-1`:** Resource-constrained e2-medium cloud virtual machine in Belgium running System 2 and System 3.
- **GCP Cloud Microservices:** Cloud Run instances, 5 Google Cloud Pub/Sub topics, Cloud Storage buckets, and an intermediate signal relay.

This distributed setup caused high operational friction: network drops stalled order execution, cross-system latency degraded fill quality, uncommitted state divergence threatened data integrity, and monthly GCP bills accumulated. On 2026-10-08, the owner authorized **Decision D1**: consolidate all three systems onto a single high-performance local computer (**Computer 4**: AMD Ryzen 7 7700, 32GB DDR5 RAM, NVIDIA RTX 5070 12GB GPU, 2TB NVMe SSD; native Windows 11 Home) and permanently shut down GCP.

2. The Chronological Cutover Sequence

Date	Phase	Actions & Deliverables	Outcome / Result
2026-10-08	Phase 0: Rescue	Rescued `src/ams/state/machine.py` (System 3 state machine) which was git-ignored on the VM and missing from all 11 Git branches. Verified 415 MB archive.	ZERO CODE LOSS
2026-10-08	Phase 1: Bootstrap	Configured Windows 11 Home: disabled sleep/hibernation, enabled NTFS long paths, installed OpenSSH, configured Tailscale VPN, added Defender exclusions.	HOST HARDENED
2026-10-08	Phase 2: Provision	Installed Python 3.12.10, PostgreSQL 16.15, and 3 separate virtualenvs (`venv-s1` , `venv-s2` , `venv-s3`) to protect pandas version boundaries (2.3 vs 3.0).	51/51 CHECKS PASS
2026-10-09	Phase 3: Acceptance	Tested unattended reboot recovery (SSH reachable in 65s); proved plain Postgres restoration; conducted R1 file-replace saturation test.	GATE PASSED

Date	Phase	Actions & Deliverables	Outcome / Result
2026-10-14	Setup Tracks	Built `sb_service.ps1` supervisor; scheduled automated encrypted daily 7z backups; established watchdog alert escalation.	SERVICES LIVE
2026-10-16	Cutover & Decommission	Retired Computer 1 crons; halted VM `trading-1`; final bucket download; resumed S2 on Computer 4; permanently deleted all GCP cloud resources.	GCP TERMINATED

3. Key Technical Challenges Solved

A. Windows Mandatory File Locking (Risk R1)

On Linux, `os.replace()` atomically overwrites open files. On Windows NTFS, any process holding an open file handle causes `os.replace()` to fail with `PermissionError: [WinError 5]`. In testing, naive file replacement lost 1,870 of 2,000 writes under saturation.

> **Resolution:** Engineered `platform.atomic` using Windows NT low-level system calls (`SetFileInformationByHandle` with `FILE_RENAME_FLAG_POSIX_SEMANTICS` and `FILE_SHARE_DELETE`). Tested over **390,000 continuous writes and 59.6 million reads with 0 failures and 0 retries**.

B. TimescaleDB Removal & PostgreSQL 16 Restoration (Risk R2)

Computer 1 stored 14.3 million ticks across 266 TimescaleDB chunks. Running proprietary extensions on Windows was deemed an unacceptable operational hazard. The database was transformed into plain PostgreSQL 16 declarative partitions, restored on Computer 4, and audited:

- Total restored rows: **21,712,916 across 24 tables**.
- Row difference against source: **0 differences (100.0% row-for-row match)**.
- Restore time: 85 seconds via `pg_restore -j 4`. Storage footprint: 5.1 GB (+16.7% vs compressed TimescaleDB).

C. Dual-Tree Monorepo & Deployment Gate

The code repository was restructured into a single monorepo deployed under a strict dual-tree workflow:

- `C:\SB\prod epo` : Production directory sitting on a detached HEAD at the pinned commit. **No manual edits or commits are ever permitted in prod.**
- `C:\SB\dev` : Active development Git worktree sharing the object store with prod.
- `deploy.ps1` Gate: Automated script that validates test suites and contract conformance on a temporary copy, gracefully stops services, updates `pins.json`, launches services, verifies health endpoints, and rolls back automatically if health checks fail.

4. GCP Resources Terminated on 2026-10-16

The following cloud assets were permanently deleted following owner verification:

- **Cloud Run:** `scalable-brain` and `telemetry-dashboard` services deleted.
- **Pub/Sub:** 5 topics and 1 subscription deleted.
- **Cloud Storage:** 4 buckets deleted (473 artifact files and 52 S3 files checksum-verified locally).
- **Compute Engine:** VM `trading-1` and 30 GB disk deleted (no snapshot retained, per owner decision).
- **Networking & Billing:** NAT router deleted; billing unlinked from account `01F047-86E4B3-8B7196`; project `scalable-brain` marked `DELETE_REQUESTED`.